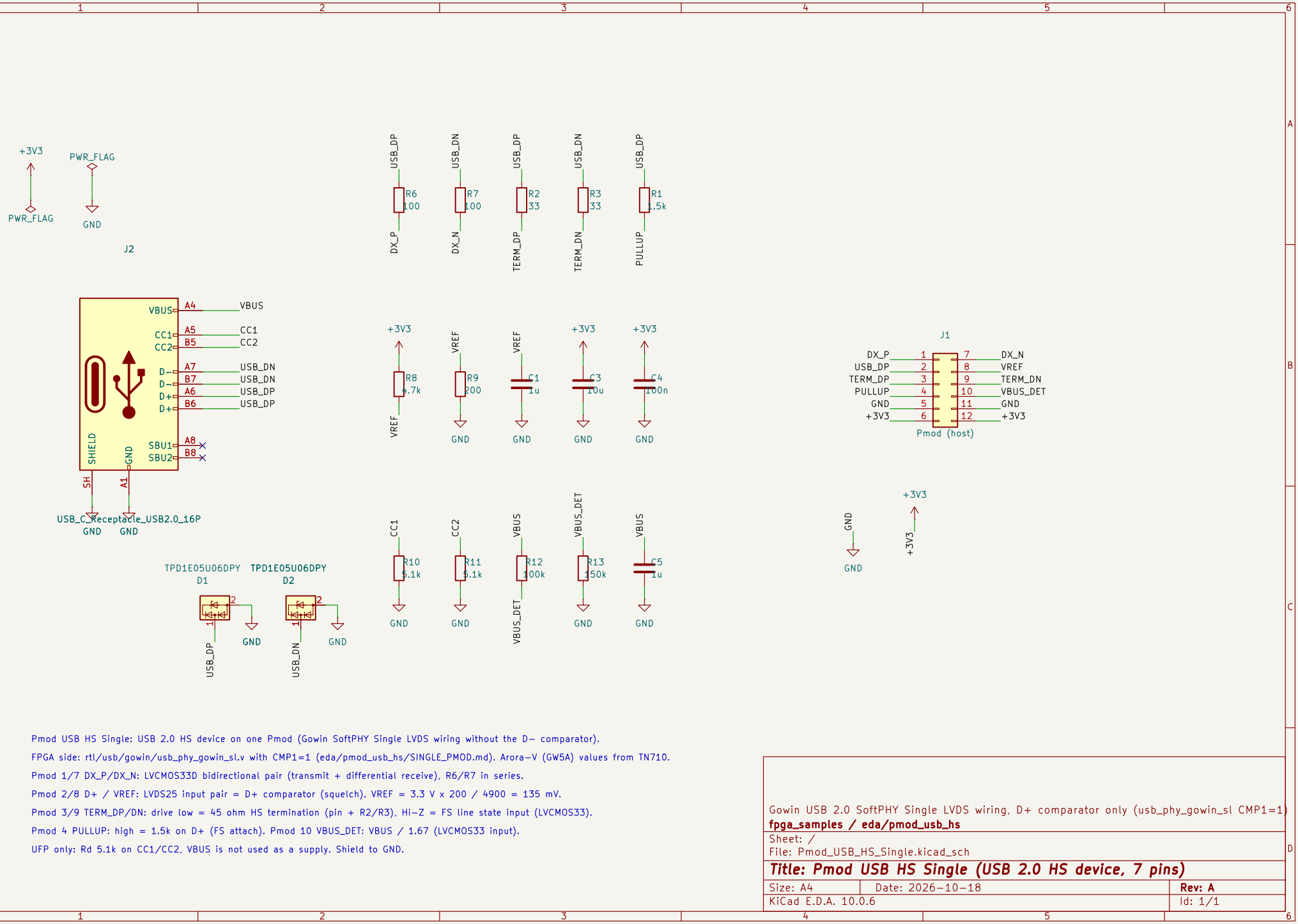

+3V3
PWR_FLAG
PWR_FLAG
GND
J2
VBUS
CC1
CC2
D−
D−
D+
D+
SHIELD
GND
SBU1
SBU2
A4
A5
B5
A7
B7
A6
B6
A8
B8
SH
A1
VBUS
CC1
CC2
USB_DN
USB_DN
USB_DP
USB_DP
USB_C_Receptacle_USB2.0_16P
GND
GND
TPD1E05U06DPY
D1
TPD1E05U06DPY
D2
USB_DP
GND
USB_DN
GND
USB_DP
USB_DN
USB_DP
USB_DN
USB_DP
R6
100
R7
100
R2
33
R3
33
R1
1.5k
DX_P
DX_N
TERM_DP
TERM_DN
PULLUP
+3V3
VREF
VREF
+3V3
+3V3
R8
4.7k
R9
200
C1
1u
C3
10u
C4
100n
VREF
GND
GND
GND
GND
CC1
CC2
VBUS
VBUS_DET
VBUS
R10
5.1k
R11
5.1k
R12
100k
R13
150k
C5
1u
GND
GND
VBUS_DET
GND
GND
J1
DX_P
USB_DP
TERM_DP
PULLUP
GND
+3V3
1
2
3
4
5
6
7
8
9
10
11
12
DX_N
VREF
TERM_DN
VBUS_DET
GND
+3V3
Pmod (host)
GND
+3V3
+3V3
Pmod USB HS Single: USB 2.0 HS device on one Pmod (Gowin SoftPHY Single LVDS wiring without the D− comparator).
FPGA side: rtl/usb/gowin/usb_phy_gowin_sl.v with CMP1=1 (eda/pmod_usb_hs/SINGLE_PMOD.md). Arora−V (GW5A) values from TN710.
Pmod 1/7 DX_P/DX_N: LVCMOS33D bidirectional pair (transmit + differential receive), R6/R7 in series.
Pmod 2/8 D+ / VREF: LVDS25 input pair = D+ comparator (squelch). VREF = 3.3 V x 200 / 4900 = 135 mV.
Pmod 3/9 TERM_DP/DN: drive low = 45 ohm HS termination (pin + R2/R3), Hi−Z = FS line state input (LVCMOS33).
Pmod 4 PULLUP: high = 1.5k on D+ (FS attach). Pmod 10 VBUS_DET: VBUS / 1.67 (LVCMOS33 input).
UFP only: Rd 5.1k on CC1/CC2, VBUS is not used as a supply. Shield to GND.
Gowin USB 2.0 SoftPHY Single LVDS wiring, D+ comparator only (usb_phy_gowin_sl CMP1=1)
fpga_samples / eda/pmod_usb_hs
Sheet: /
File: Pmod_USB_HS_Single.kicad_sch
Title: Pmod USB HS Single (USB 2.0 HS device, 7 pins)
Size: A4
Date: 2026−10−18
Rev: A
KiCad E.D.A. 10.0.6
Id: 1/1